

Welcome to Section! Week 5



Slides by Cameron Mohne and Maggie Lee

Agenda



1.

Check-in



2.

Recap



3.

Section Problem:
Random Circles



Questions?

Before we jump into the recap, are there any questions?



Lecture Recap



Recap - Functions 2.0





Functions 2.0



Parameters

Parameters are a way to pass data to your helper functions. Variables made in a function are invisible to other functions unless it is passed through a parameter.



```
def function_one():  
    variable_one = 0  
    function_two()
```

Incorrect

Not possible! *function_two* doesn't know that *variable_one* exists!

```
def function_two():  
    variable_two = variable_one
```

```
def function_one():  
    variable_one = 0  
    function_two(variable_one)
```

Correct

We use a parameter to pass the data instead

```
def function_two(example_param):  
    variable_two = example_param
```



Functions 2.0



Returns

Similar to parameters, but in the reverse direction! We use return statements to pass data from a function to wherever it was called.



One example is the `input()` function that you've been using! Whenever you have a function that returns something, you usually want a variable to store that data!



```
def function_one():  
    name = input("Enter your name: ")  
  
def input(prompt):  
    ... # Some code  
    return user_input
```

**`input()`
returns data
for you to use,
wherever you
called it from!**

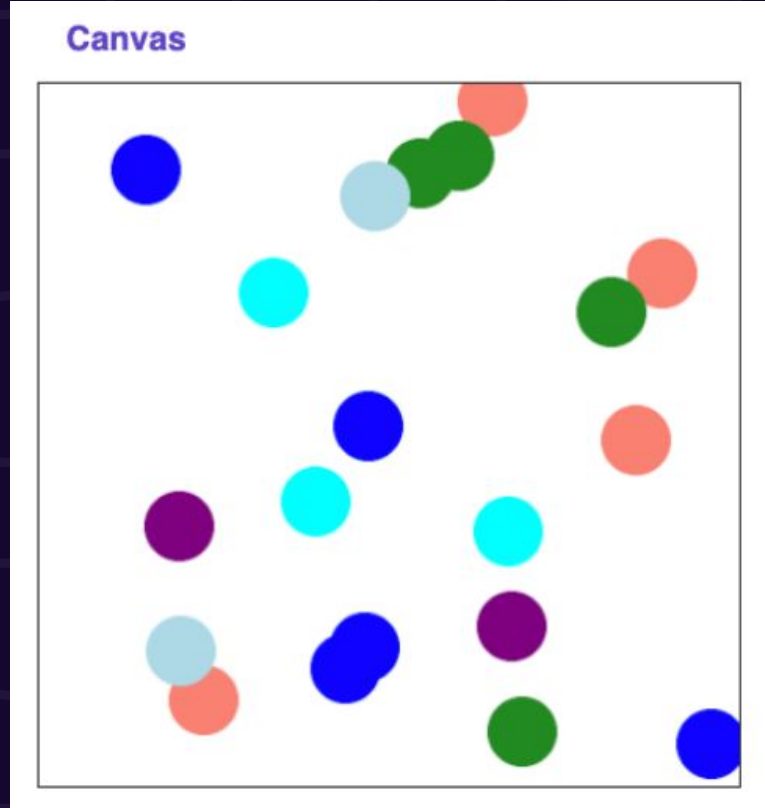


Recap - Graphics



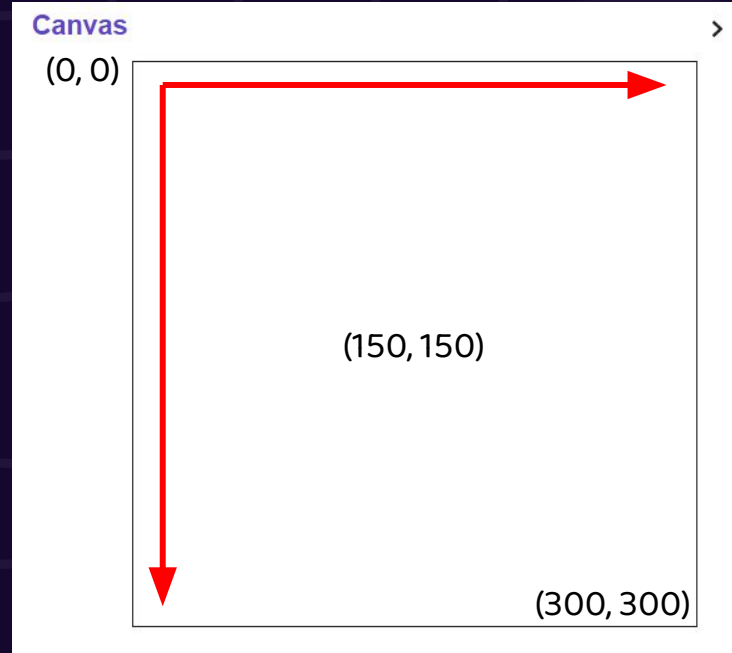
Canvas

The canvas is a fun way for us to visualize and interact with our code in a somewhat similar way to Karel!



Canvas

Remember how pixels are laid out: starting at (0, 0) for the top left and increasing in x/y-values as you go right/down respectively



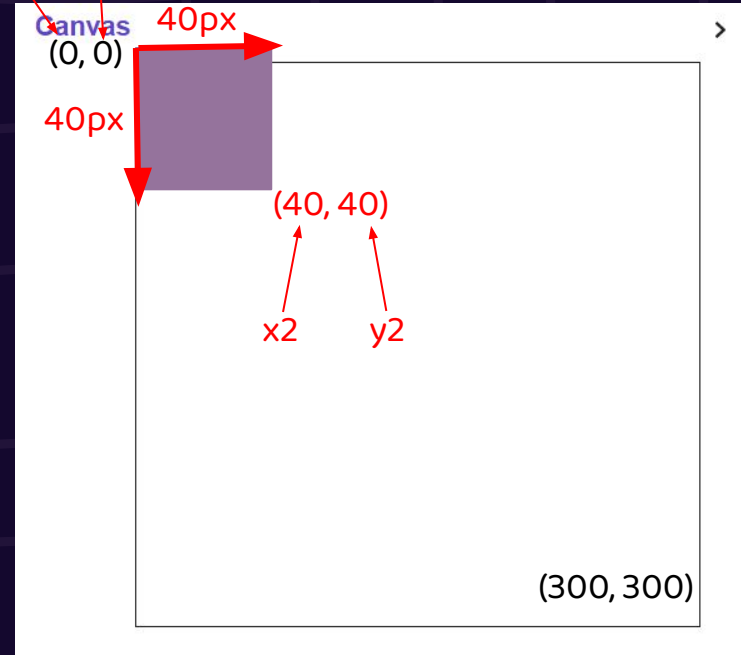
Canvas

To draw something, we need to use four parameters. These numbers represent the coordinates of the two points used to draw the image. Since the function is a part of our canvas, we use our canvas followed by a period and then the function name. We can also add an extra parameter for color if we want.

Example:

```
canvas.create_rectangle(0, 0, 40, 40, "Purple")
```

x1 y1 x2 y2 color





Recap Questions

Wow! That was a lot! You've also been learning a lot.
Are there any questions over lecture content or anything else?



Section Problem: Random Circles



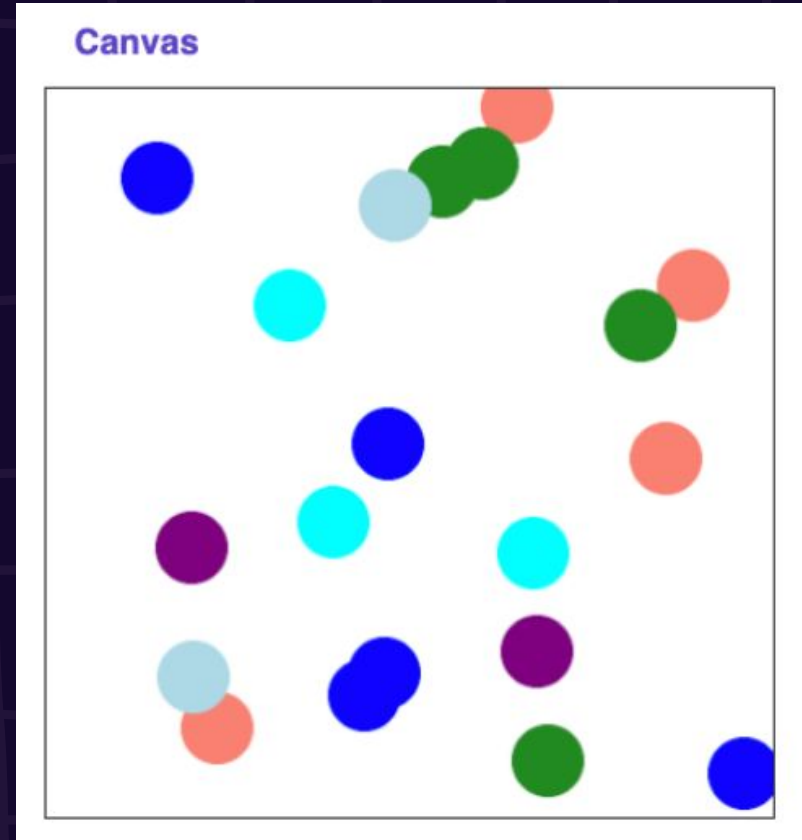


Random Circles

As an intro to functions & graphics, we want to get comfortable with:

- 1) The concepts of drawing and indexing with pixels
- 2) Calling functions with parameters/returns.

To practice, let's write a program that **draws 20 circles at random positions with random colors** on the canvas.

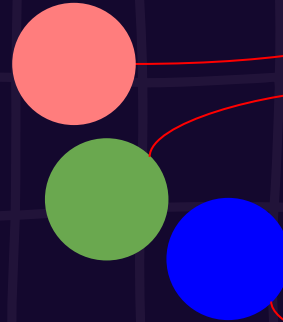




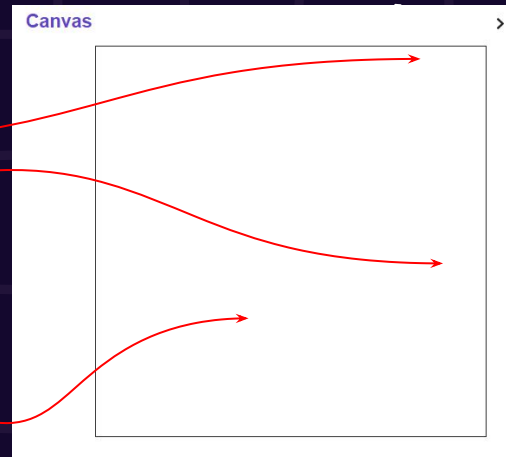
x20

End Goal

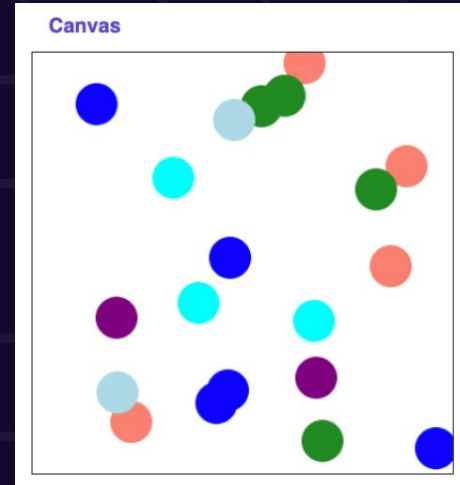
Our program takes in a blank canvas and then **draws 20 circles at random positions with random colors** on the canvas.



Example Starting Canvas



Example Output Canvas





Problem Details



Canvas

The canvas is created for you already with some functions provided

Canvas Details

Canvas width/height are provided as global variables

Colors

We provide a function that returns a random color

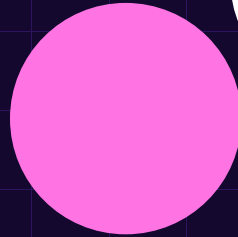
Circle Details

We give you the diameter size of the circles and the number of circles to make



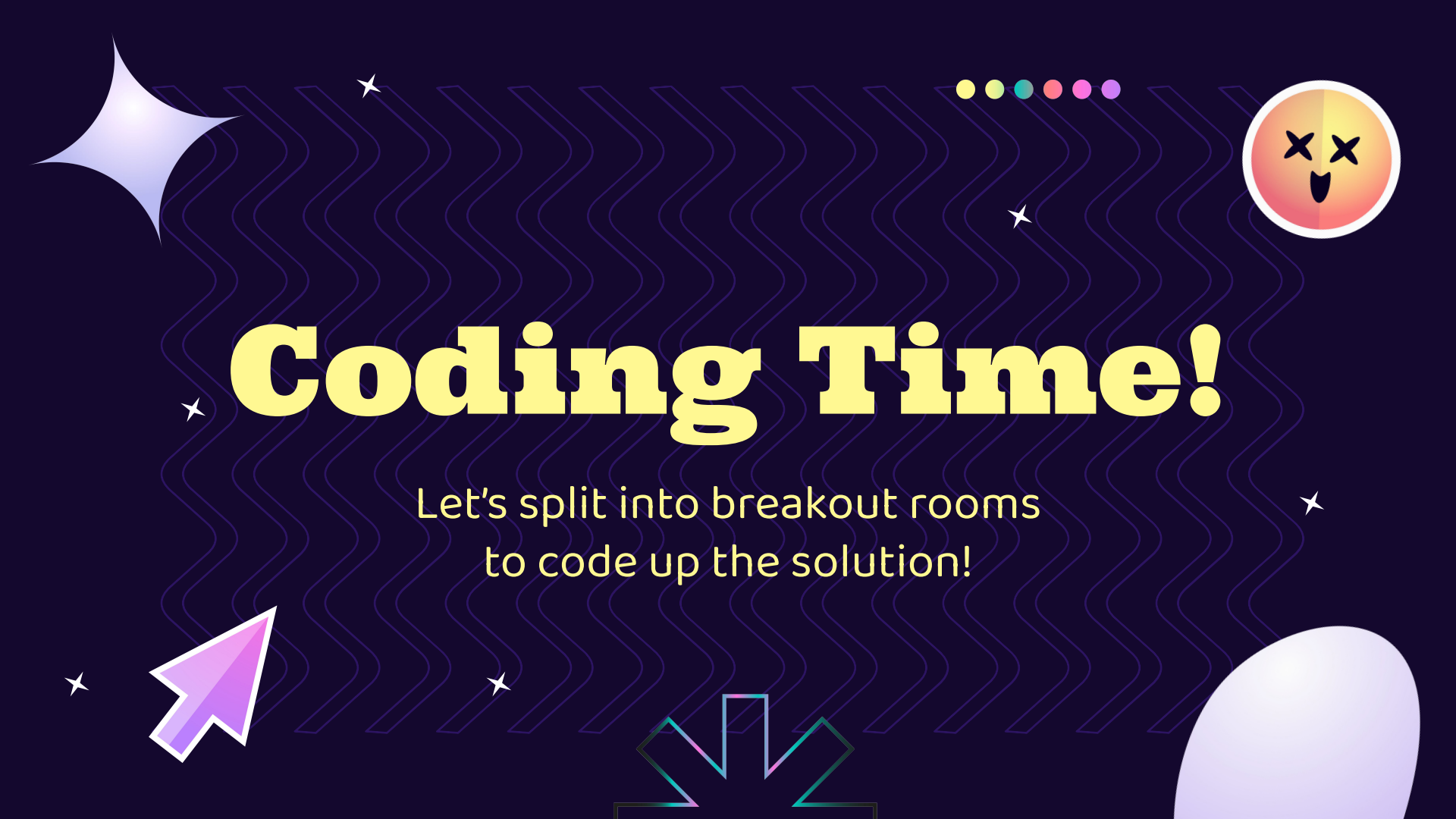
Pre-code Discussion

Say we want to make a single circle, how would we do that?



Hi, I'm a
circle! And
single!

Are there any questions regarding the canvas,
graphics, drawing, or anything else?



Coding Time!

Let's split into breakout rooms
to code up the solution!



Post-code Discussion

Are there any questions at all?

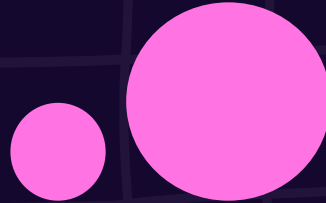


Extensions



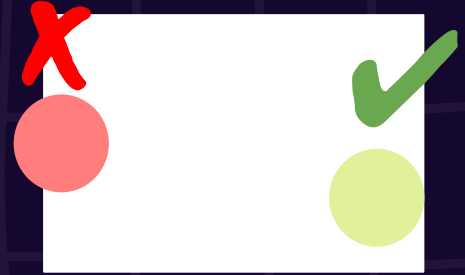
Number of Circles

Create a random amount of circles instead of a set amount!



Size of Circles

Have the circles be of random size instead of a set size!



Bounds of Circles

Make sure all circles are in the bounds of the canvas!

